

A. The River System

- 1 River Systems and classification
- 2 Catchment Hydrology, Rainfall-Runoff, Climate
- 3 River Basin Hydrology: Rainfall to River Flow

B. River management, functions and interventions

- 1 National and International River Management
- 2 Flood Risk Management
- 3 Inland Navigation
- 4 Nature-Based Solutions and River Restoration
- 5 River Pollutants and Plastics
- 6 Drought Risk Management
- 7 River Ecology
- 8 Dams, Hydropower, and Reservoirs

C. Flow and Transport

- 1 Normal Flow
- 2 Backwater Curves
- 3 Flood Waves
- 4 Sediment Production and Suspended Load transport
- 5 Bed load transport
- 6 Transport and Dispersion
- 7 Ice river dynamics

D. Morphological Response to Perturbations

- 1 Initial Channel Response
- 2 Long-term Channel Response (constant)
- 3 Long-term Channel Response (variable)
- 4 Transient Channel Response

E. Real world cases

- 1 July 2021 Peak Flow Meuse
- 2 Drought of 2026 in Europe
- 3 TO BE DEFINED
- 4 TO BE DEFINED